Jean Monnet University

"Data Analysis" Exam
MLDM and iPSRS Master programs

December 2025 - **(2 hours)**

Pocket calculators and two 2-sided papers of handwritten notes are allowed.

Exercise 1 (5 points): Optimization

Given the following constrained optimization problem:

$$\min_{(x,y)\in\mathbb{R}^2} \; f(x,y) = (x-2)^2 + (y+1)^2,$$
$$\text{subject to} \quad x + 2y - 4 = 0.$$

1. Using the Hessian matrix, show that $f(x, y)$ is a convex function.
2. Find the unique minimizer of this convex optimization problem using the Lagrangian.

Exercise 2 (5 points): Likelihood Maximization

Suppose that $X_1, X_2, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables following a geometric distribution with unknown success probability $p \in (0, 1)$. This distribution gives the probability that the first occurrence of success requires x_i independent trials. It is defined as follows:

$$P(X = x_i) = p(1-p)^{x_i-1}, \quad x_i = 1, 2, 3, \ldots$$

1. Compute the likelihood function $L(p)$ based on the sample $(X_1, X_2, \ldots, X_n)$ and deduce the log-likelihood function $\ell(p) = \log L(p)$.
2. Using the likelihood maximization principle, find the estimate $\hat{p}$ of the theoretical parameter p.

Exercise 3 (6 points): PCA

We consider three centered observations in $\mathbb{R}^2$:

$$x^{(1)} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad x^{(2)} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \quad x^{(3)} = \begin{pmatrix} -2 \\ -2 \end{pmatrix}.$$

1. Compute the covariance matrix Σ.
2. Find the two **eigenvalues** of Σ.
3. Compute the unit **eigenvector** corresponding to the largest eigenvalue.
4. Compute the proportion of explained variance by a projection onto $\mathbb{R}$.
5. Plot the projected points according to the first principal component.

Exercise 4 (4 points): Multiple Choice Questions

Write the answers to the following questions on your answer sheet (**only one is correct**). Each correct answer **adds 1/2**. Each incorrect answer **subtracts 1/4**.

1. If two rows of a square matrix A are identical, then:
   (a) $\det(A) = 0$
   (b) $\det(A) = 1$
   (c) $\det(A)$ is the determinant of the submatrix without these two rows.

2. Suppose A is a 2×2 matrix with eigenvalues λ_1 and λ_2. The trace of A is:
   (a) $\lambda_1^2 + \lambda_2^2$
   (b) $\lambda_1 + \lambda_2$
   (c) $\lambda_1 \lambda_2$

3. If A is an orthogonal matrix, then:
   (a) $A^T A = 0$
   (b) $A^{-1} = -A$
   (c) $A^T A = I$

4. Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. Compute $\det(A)$.
   (a) 5
   (b) 6
   (c) 3

5. Let $A = \begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix}$. Which of the following is an eigenvector associated with the eigenvalue -2?
   (a) $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
   (b) $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
   (c) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

6. The inverse of $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ is:
   (a) $\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$
   (b) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$
   (c) $\begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$

7. Given the gaussian variable $Y \sim N(10, 2)$ of mean 10 and standard deviation 2. $P(Y \leq 12)$ is equal to:
   (a) 0.8413
   (b) 0.1587
   (c) 0.5

8. If $Z \sim N(0, 1)$, which of the following is true?
   (a) $P(-1 \leq Z \leq 1) \approx 68\%$
   (b) $P(-1 \leq Z \leq 1) \approx 95\%$
   (c) $P(-1 \leq Z \leq 1) \approx 99\%$